

# NITISH KUMAR

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## SKILLS

|                   |                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------|
| Languages:        | Python, C#, Julia, SQL, R                                                                                      |
| AI/ML Frameworks: | scikit-learn, TensorFlow, PyTorch, XGBoost, Optuna                                                             |
| Dev & Deployment: | FastAPI, Docker, Git, MLflow, Jupyter Notebooks, pybind11                                                      |
| Big Data:         | Apache Hadoop, Apache Hive, Linux                                                                              |
| Other Tools:      | .NET, OpenCV, Unity, Streamlit, MS SQL Server                                                                  |
| Technical:        | Data Wrangling, Feature Engineering, Model Evaluation, Statistical Modelling, API Integration, Pose Estimation |
| Soft Skills:      | Adaptability, Problem Solving, Research Communication, Cross-functional Collaboration, Attention to Detail     |

## EXPERIENCE

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| <b>Academy Analyst</b>   Python, Video Analysis, Tactical Reporting                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Aug 2024 – Dec 2024 |
| <i>Global Sports Coaching Academy, New Delhi</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                     |
| <ul style="list-style-type: none"><li>Integrated structured match and training data to deliver performance insights for coaching staff.</li><li>Developed video and data analysis pipelines to support tactical planning and post-match reviews.</li><li>Presented findings to coaches, contributing to improved team consistency and a youth tournament runner-up finish.</li><li>Identified development trends in player fitness and technical execution through qualitative and quantitative reporting.</li></ul> |                     |

## PROJECTS

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| <b>Kinesis — AI Biomechanics Platform</b>   FastAPI, MediaPipe, ViTPose, React Native                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Apr 2026 – Present  |
| <ul style="list-style-type: none"><li>Built a single-camera markerless motion capture platform targeting football academies — a low-cost alternative to £50k+ lab setups.</li><li>Engineered a core pipeline from pose estimation (MediaPipe/ViTPose) through biomechanical feature extraction to injury risk and performance scoring.</li><li>Architected a FastAPI + Python ML backend, React Native mobile app, and ASP.NET Core admin panel.</li><li>Pursuing university research partnerships and pilot deployments with grassroots football clubs.</li></ul> |                     |
| <b>J1 League Player Performance Dashboard</b>   Python, Streamlit, StatsBomb, K-Means                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | May 2026            |
| <ul style="list-style-type: none"><li>Developed an interactive Streamlit dashboard for scouting and analysing J1 League player performance using StatsBomb and Hudl data.</li><li>Applied K-Means clustering to segment player profiles and Ridge regression for performance modelling across positional groups.</li><li>Implemented cosine similarity-based player comparison to identify stylistically similar players across the league.</li></ul>                                                                                                              |                     |
| <b>ML-based Pitch Recommendation System</b>   Python, XGBoost, Pandas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Sep 2025 – Oct 2025 |
| <ul style="list-style-type: none"><li>Trained an XGBoost model on 3 MLB seasons of play-by-play data to recommend optimal pitch types per matchup.</li><li>Engineered features from count leverage, batter tendencies, and pitcher profiles with emphasis on interpretability and sports-strategy alignment.</li></ul>                                                                                                                                                                                                                                             |                     |
| <b>Timeline Scout — Video Tagging Tool</b>   C#, .NET, OpenCV                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Feb 2025 – May 2025 |
| <ul style="list-style-type: none"><li>Built a low-cost alternative to industry-standard sports video annotation tools with AI-powered automation.</li><li>Integrated computer vision modules for automatic event detection and player tracking to streamline analyst workflows.</li></ul>                                                                                                                                                                                                                                                                          |                     |
| <b>Elsword API</b>   Python, Discord API, REST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Dec 2024 – Jun 2025 |
| <ul style="list-style-type: none"><li>Developed a custom REST API to retrieve real-time game data for the MMORPG Elsword, serving as a backend for a Discord guide bot.</li><li>Implemented endpoints for character stats, equipment lookups, and game metadata to support community tooling.</li></ul>                                                                                                                                                                                                                                                            |                     |
| <b>Baseball Pitch Type Classifier</b>   Python, Optuna, scikit-learn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Sep 2024 – Oct 2024 |
| <ul style="list-style-type: none"><li>Trained a pitch type classification model on MLB Statcast data, optimised via Optuna hyperparameter tuning.</li><li>Evaluated multiple classifier architectures to maximise accuracy across pitch categories including fastballs, breaking balls, and off-speed pitches.</li></ul>                                                                                                                                                                                                                                           |                     |

## PUBLICATIONS

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| <b>Digital Fencing — A Solution to Animal-Human Conflict</b>                                                                                                                                                                                                                      | Jun 2024 |
| <i>Rohit Beniwal, Nitish Kumar, Parshant Niteesh Rathore · IEEE ICAC3N 2023</i>                                                                                                                                                                                                   |          |
| <ul style="list-style-type: none"><li>Co-authored and presented at IEEE ICAC3N 2023. Contributed data analysis, modelling, and system simulation design.</li><li>DOI: <a href="https://doi.org/10.1109/ICAC3N60023.2023.10541347">10.1109/ICAC3N60023.2023.10541347</a></li></ul> |          |

## ACHIEVEMENTS & CO-CURRICULAR

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|-----------------------------------------------------------------------------------------------|----------|
| Top 15 — <b>ReSurgeance under Cognitia 2026 Hackathon</b>   Lovely Professional University.   | Mar 2026 |
| Represented hostel in <b>Inter-hostel Basketball</b>   Lovely Professional University.        | Feb 2026 |
| Represented department in <b>Inter-department Football</b>   Lovely Professional University.  | Feb 2026 |
| Top 15 — <b>Smart India Hackathon 2025 (Internal Round)</b>   Lovely Professional University. | Nov 2025 |

## EDUCATION

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|-------------------------------------------------|---------------------|
| <b>Lovely Professional University</b>           | Aug 2025 – Present  |
| M.Sc. Statistics and Data Analytics, CGPA: 7.25 | Phagwara, Punjab    |
| <b>Delhi Technological University</b>           | Aug 2019 – Jun 2023 |
| B.Tech in Computer Engineering, CGPA: 6.81      | New Delhi           |